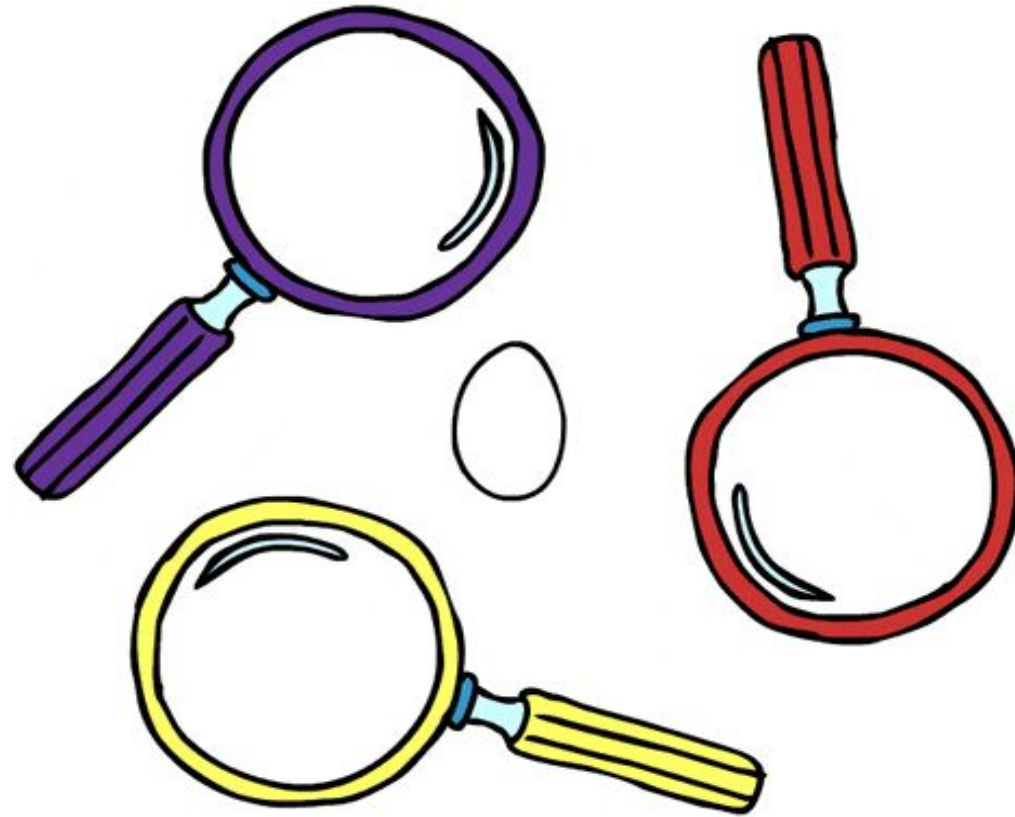


Research Methodology-II

Qualitative Research



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Objectives

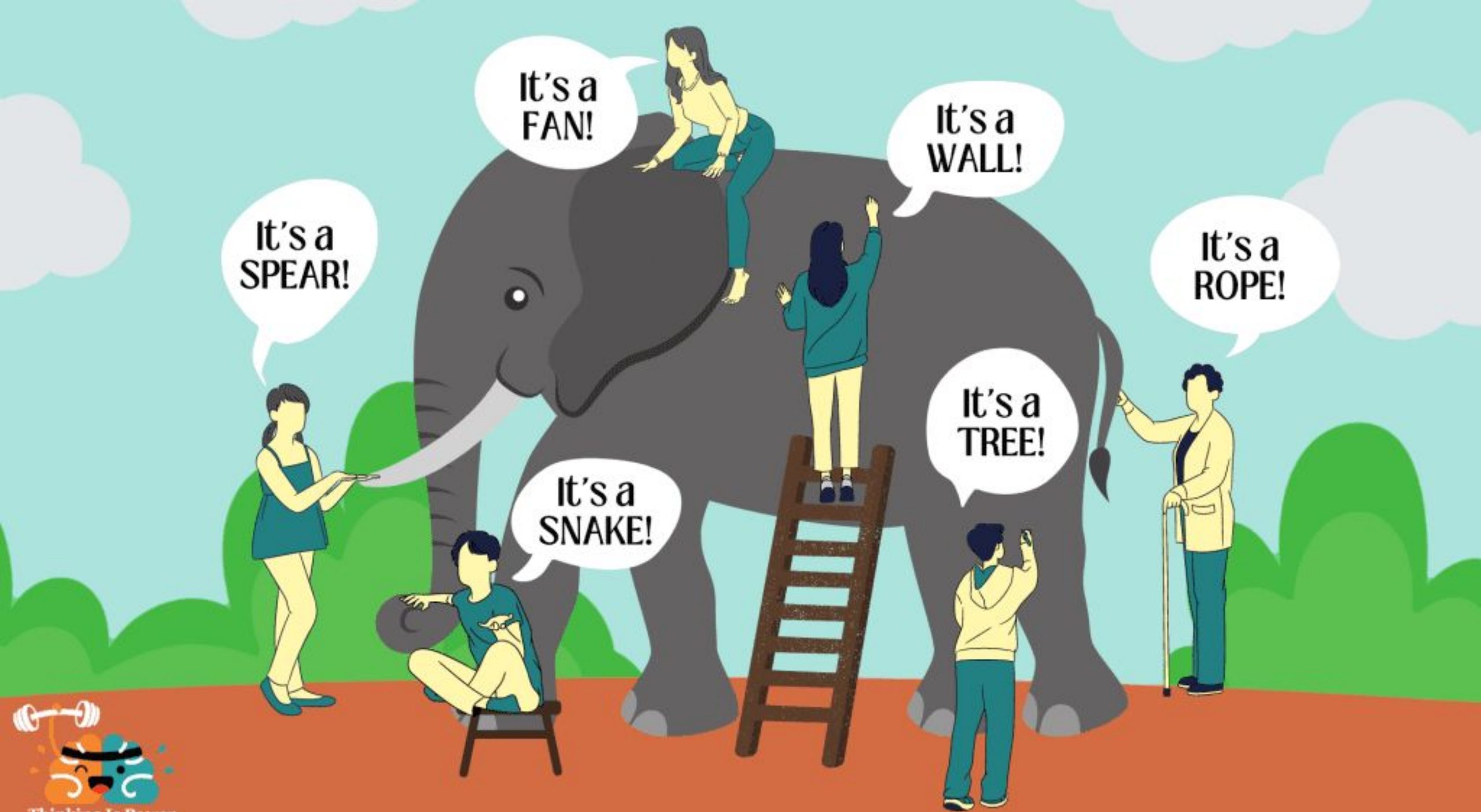


At the end of the lecture student should be able to:

- Define qualitative research and understand its purpose in health and dental sciences.
- Differentiate between qualitative and quantitative research approaches.
- Identify key characteristics of qualitative research, such as its focus on experiences, perceptions, and meanings.
- Describe common qualitative research methods, including interviews, focus groups, and observations.

Qualitative Research

- Qualitative research is a method of inquiry that **focuses on understanding human experiences, behaviors, emotions, and perceptions through non-numerical data** such as interviews, observations, and open-ended questions.
- It aims to explore **the "why" and "how"** behind certain phenomena rather than just measuring "how many" or "how much."



It's a
FAN!

It's a
WALL!

It's a
ROPE!

It's a
TREE!

It's a
SNAKE!

It's a
SPEAR!

Qualitative Research: Why???



- **Understanding Patient Perspectives:** It helps uncover patients' beliefs, fears, motivations, and experiences related to oral health, treatment, and dental care

Example: How do patients describe their emotional experiences during a dental visit?

- **Exploring Health Behaviors:** Qualitative research provides insight into why people adopt or avoid certain dental hygiene practices or treatments.

Example: What are the reasons why some individuals avoid regular toothbrushing despite being aware of its benefits?

- **Improving Communication:** It aids in developing better patient-dentist communication strategies by understanding how patients perceive health messages or dental visits.

Example: How do patients perceive the clarity and effectiveness of communication from their dentists during treatment planning?

Qualitative Research: Why???



- **Developing Patient-Centered Care:** Findings can guide the design of care that is more personalized, empathetic, and responsive to individual needs.

Example: What are patients' expectations and preferences for personalized dental care in a clinical setting?

- **Identifying Barriers to Care:** It helps explore social, cultural, and economic factors that prevent patients from accessing or adhering to dental care.

Example: What social and cultural factors do low-income patients perceive as barriers to accessing routine dental services?

- **Informing Health Policy and Education:** Qualitative insights are valuable in shaping dental education curricula, public health campaigns, and policy making by addressing real-world patient concerns and experiences.

Example: How do dental students perceive the effectiveness of their curriculum in preparing them for real-world patient interactions?

QUANTITATIVE RESEARCH



QUALITATIVE RESEARCH



Feature	Qualitative Research	Quantitative Research
Nature of Data		
Purpose		
Research Question		
Data Collection Methods		
Data Analysis		
Outcome		
Sample Size		
Role of Researcher		
Use in Dental Sciences		



A top-down illustration of various business tools and collaborative elements. It includes a laptop with hands typing, a smartphone being held, a tablet with a hand pointing, a calculator, a coffee cup, a pair of glasses, a color calibration chart, a pen, a USB cable, a document with a checklist, and a hand holding a pen over a document with a pie chart. The items are arranged in a circular pattern, suggesting a central point of collaboration or a shared workspace.

Qualitative Research: Attributes



- **Contextual Understanding:**

Emphasizes understanding behavior or experiences within the social, cultural, or personal context.

- **Subjective Interpretation:**

The researcher plays an active role in interpreting meanings and identifying themes in the data.

- **Flexible and Adaptive:**

The research process may evolve as new insights emerge during data collection and analysis.

Qualitative Research: Attributes



- **Thematic Analysis:**

Data is analyzed by identifying recurring themes, patterns, or concepts rather than using statistical tests.

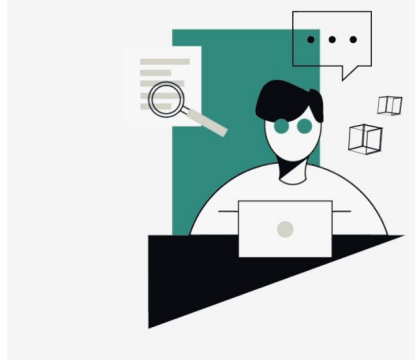
- **Participant-Centered:**

Gives voice to the perspectives, experiences, and emotions of participants.

- **Natural Settings:**

Often conducted in real-life environments (e.g., clinics, communities) rather than controlled laboratories.

Qualitative Research: Methods



- Interviews

- One-on-one conversations to explore participants' views and experiences.
- Structured, semi-structured, or unstructured.
- Ideal for exploring individual perspectives in depth.

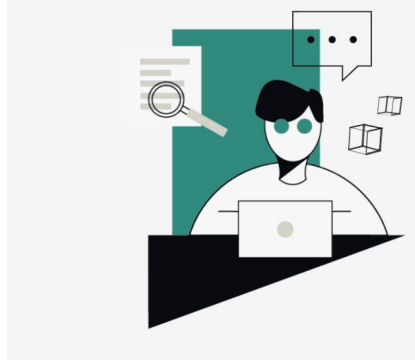
Advantages

- Allows deep exploration of personal experiences and insights
- Flexible—interviewer can follow up on interesting points
- Good for sensitive topics in private settings

Disadvantages

- Time-consuming to conduct
- Risk of interviewer bias or leading questions
- Relies heavily on participants' ability to articulate their thoughts

Qualitative Research: Methods



- Focus Groups

- Group discussions guided by a moderator to explore shared experiences and attitudes.
- Typically 6–10 people.
- Useful for understanding group norms, interactions, and collective opinions.

Advantages

Encourages interaction and idea sharing among participants

Can generate a wide range of views in a short time

Efficient for collecting data from multiple people simultaneously

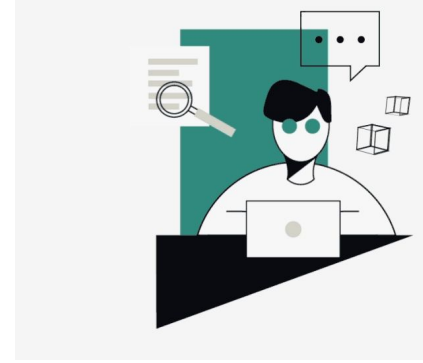
Disadvantages

Dominant voices may overshadow others

Participants may be reluctant to share sensitive information in a group

Group dynamics may influence individual responses (e.g., peer pressure)

Qualitative Research: Methods



- Observations

- Watching and recording behaviors and interactions in natural settings.
- Participant or non-participant.
- Effective for studying real-life settings and social practices.

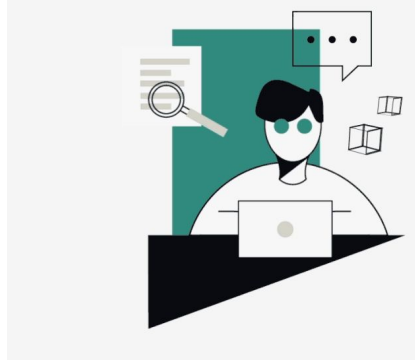
Advantages

- Captures actual behavior in real-life settings
- Provides context-rich and non-verbal data
- Useful when participants cannot clearly express experiences verbally

Disadvantages

- Observer presence may influence behavior (Hawthorne effect)
- Can be time-consuming and labor-intensive
- Interpretation may be subjective or biased

Qualitative Research: Methods



- Case Studies

- In-depth analysis of a single case (e.g., a person, group, event, or organization).
- Useful for gaining detailed contextual insights and understanding complex issues.

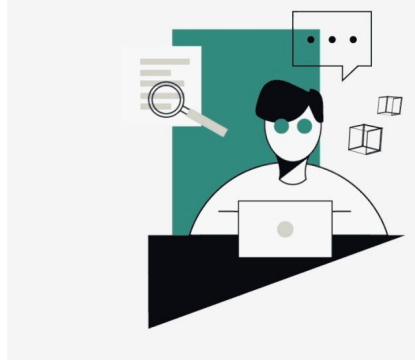
Advantages

- Offers deep, detailed understanding of complex phenomena
- Useful for exploring rare or unique situations
- Allows multiple data sources (interviews, documents, observations)

Disadvantages

- Limited generalizability to other contexts
- Can be time-consuming to collect and analyze large volumes of data
- Risk of over emphasis on one case

Qualitative Research: Methods



- Document or Content Analysis

- Systematic analysis of text-based or visual materials (e.g., reports, social media posts, diaries).
- Helps uncover underlying themes or patterns in communication

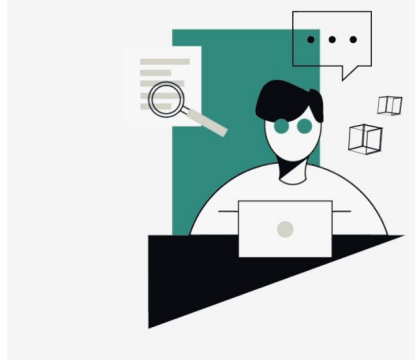
Advantages

- Useful for historical or retrospective analysis
- Can be unobtrusive—does not require direct interaction with participants
- Enables analysis of trends over time

Disadvantages

- Limited to available or accessible documents
- May lack context or explanation behind content
- Interpretation may be subjective without triangulation

Qualitative Research: Methods



- Ethnography

- Extended study of people and cultures in their natural environment, often involving immersion.
- Common in anthropology and sociology to understand cultural practices and beliefs.

Advantages

Provides deep cultural insights through long-term immersion

Captures practices and meanings in real-life contexts

Offers holistic understanding of a group or community

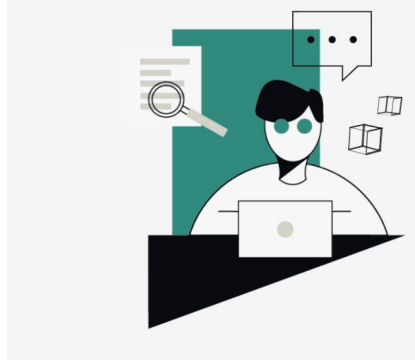
Disadvantages

Requires significant time and researcher involvement

Researcher may "go native" and lose objectivity

Ethical issues related to informed consent and privacy

Qualitative Research: Methods



- Narrative Research

- Examines personal stories to understand how individuals make sense of their experiences.
- Helps reveal how people construct meaning through storytelling.

Advantages

Reveals how individuals make meaning of their experiences through storytelling

Emphasizes individual voices and perspectives

Useful in understanding identity, life transitions, and cultural context

Disadvantages

Time-intensive to collect and analyze narratives

Findings may be difficult to generalize

May be influenced by memory recall or storytelling style



- **What is the primary focus of qualitative research?**

- A. Measuring numerical outcomes
- B. Understanding human experiences and perceptions
- C. Testing hypotheses using statistics
- D. Conducting randomized trials



Answer: B

- **Which of the following is NOT a common method of qualitative research?**

- A. Case study
- B. Focus group
- C. Laboratory experiment
- D. Ethnography



Answer: C

- **Qualitative research is particularly useful for studying:**

- A. Drug efficacy
- B. Behavior in natural settings
- C. Genetic mutations
- D. Disease prevalence

 Answer: B

- **In qualitative research, the role of the researcher is:**

- A. Passive observer
- B. Completely objective
- C. Active interpreter
- D. Data processor



Answer: C

- **Which qualitative method involves analyzing personal stories?**

A. Ethnography

B. Narrative research

C. Focus group

D. Document analysis



Answer: B

- **A focus group typically includes how many participants?**

A. 2–4

B. 6–10

C. 15–20

D. Over 30



Answer: B

- **What type of sampling is common in qualitative research?**

- A. Random sampling
- B. Stratified sampling
- C. Purposive sampling
- D. Systematic sampling



Answer: C

- **Thematic analysis in qualitative research involves:**

- A. Statistical hypothesis testing
- B. Coding and identifying patterns in data
- C. Comparing means and medians
- D. Randomizing participants into groups



Answer: B

- **Which method is best for studying interactions in a natural setting?**

- A. Interviews
- B. Case studies
- C. Observations
- D. Surveys



Answer: C

- **Which is a key characteristic of qualitative research?**

- A. Use of control groups
- B. Emphasis on objectivity
- C. Use of non-numerical data
- D. Focus on generalizability



Answer: C

- **Which qualitative method is most suitable for studying cultural beliefs and practices?**

- A. Ethnography
- B. Narrative research
- C. Content analysis
- D. Case study



Answer: A

- **Document analysis is primarily concerned with:**

- A. Lab results
- B. Patient interviews
- C. Visual or text-based materials
- D. Questionnaire data



Answer: C

- **One strength of focus groups is:**

- A. Individual depth
- B. Group dynamic insights
- C. Complete anonymity
- D. High generalizability



Answer: B

- **A major weakness of observations is:**

- A. Limited real-life context
- B. Risk of Hawthorne effect
- C. Poor depth of data
- D. Lack of flexibility



Answer: B

- **Which of the following is TRUE about case studies?**

- A. They involve large random samples
- B. They provide shallow, quick insights
- C. They allow detailed, in-depth understanding
- D. They are usually experimental



Answer: C

- **In qualitative research, open-ended questions are used to:**

A. Test hypotheses

B. Control variables

C. Measure frequency of events

D. Allow participants to freely express experiences



Answer: D

- **One of the main purposes of qualitative research in dental education is to:**
 - A. Improve communication and patient-centered care
 - B. Compare drug efficacy
 - C. Understand tooth structure
 - D. Perform statistical validation



Answer: A

- **Which of the following methods is ideal for exploring an individual's personal experience in depth?**

- A. Focus group
- B. Interview
- C. Observation
- D. Survey



Answer: B

- **What is a common challenge in conducting focus groups?**

A. Lack of data

B. Inflexibility

C. Dominance of certain participants

D. No opportunity for discussion



Answer: C

- **What type of data is typically generated in qualitative research?**

A. Percentages

B. Scores

C. Narratives and themes

D. Charts and graphs



Answer: C

Scientific Method



A researcher spends six months observing tribal dental hygiene practices while living in a rural village. Research Type?

Ethnography

A dental student interviews patients one-on-one to explore their fear of root canal treatments.

Interviews

Researchers hold a group discussion with 8 patients to explore their experiences with dental implants.

Focus Group

A study examines diaries written by patients documenting their post-operative dental pain for two weeks.

Document or Content Analysis

A researcher closely observes patient behavior in a dental waiting area without interacting.

Non-Participant Observation

A professor conducts an in-depth study of one dental student's experience during their clinical rotations

Case Study

Patients narrate their journey of recovering from maxillofacial surgery

Narrative Research

What are the cultural practices influencing patient care in a rural hospital?

Ethnography

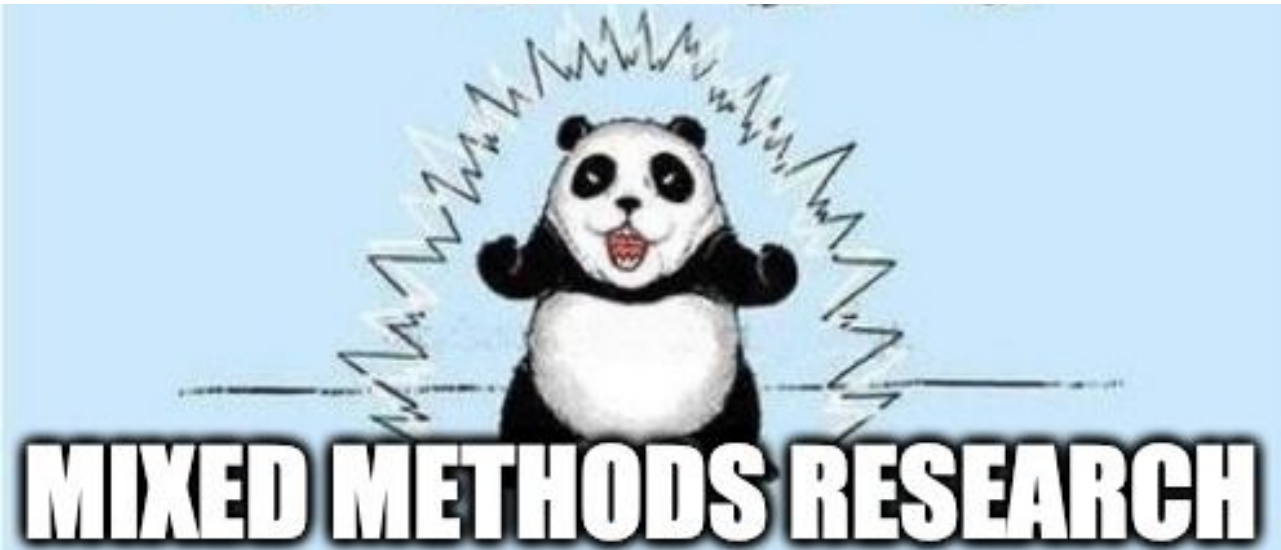
How is a new teaching strategy implemented in a medical school?

Case study

**QUANTITATIVE
RESEARCH**



**QUALITATIVE
RESEARCH**



MIXED METHODS RESEARCH

when waiting for the
boring lectures to end.



Thank You